Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/05/2024
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	Countermeasures Against Adverse Effects of Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/31/2019	**End Date:**	09/30/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2023 per NSSC information (Ed., 5/18/23) NOTE: End date changed to 5/30/2023 per NSSC information (Ed., 5/8/22) NOTE: End date changed to 5/30/2022 per NSSC information (Ed., 3/29/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0435		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This is the task book for the Phase 1 study of our project entitled “Countermeasures against adverse effects of space radiation” in which only male mice are included. The search for efficient radiation countermeasures in space is a high priority. Apigenin (AP), a natural flavone, displays potent antioxidant, anti-inflammatory, anti-microbial, and anti-neoplastic activities. We found that AP protects human lymphocytes from in vitro ?-irradiation and that intraperitoneal AP diminishes inflammation in the bone marrow (BM) of ?-irradiated mice. However, the countermeasure activity of AP has not been studied for space radiation. Here we report the efficacy of oral AP for space-radiation-induced toxicity in hematopoietic and gut tissues. We gave food with (20 mg/kg/bw) or without (0 mg/kg/bw) AP to male C57BL/6 mice (with or without irradiation) before (5 days) and after total-body exposure to 0 or 0.5 Gy of 28Si ions. There were 4 experimental groups. Mice with the AP diet (with or without irradiation) were maintained on the AP diet until mice reach 620 days of age (at which time the weights of mice in each group did not significantly change) before switching to a regular diet. At 1-week after-irradiation, 6 mice from each group were randomly euthanized to test AP countermeasure for space-radiation-induced early toxicity (hematological toxicity and inflammation) in BM cells and the duodenum (the main absorption site for oral AP) collected from the same mouse. We also evaluated the effectiveness of AP in protecting/mitigating mice from Si-induced chromosome aberrations (using the mouse blood in vivo micronucleus assay). Our data enable the evaluation of the countermeasure efficacy of AP across tissues at risk for cancer induction, i.e. bone marrow, the lung, and the gastrointestinal (GI) tract. Importantly, we used the 16S rRNA amplicon sequencing method to study the impact of the AP diet on the gut microbiota of mice with or without exposure to silicon ions. This multi-tissue of the same exposed individual approach has not been used in space research. All remaining mice were observed for morbidity and mortality until they reach about 770 days of age to determine the effectiveness of AP in counteracting Si-induced cancer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	People can be exposed to radiation in many different ways, both intentionally or unintentionally, on earth or in the space environment. Hence, the search for efficient medical countermeasures (protectors, those given before exposure; mitigators, those given after exposure) is a high priority in radiation protection on earth and space. The ideal biological countermeasure for counteracting harmful effects of radiation should be (i) capable of suppressing inflammation (one of the important physiological changes related to untoward health outcome) after radiation exposure, (ii) enabling the prevention and mitigation of radiation-induced damage across tissues, (iii) amenable to apply before the exposure, regularly, and after radiation exposure, and (iv) easily administered (i.e. oral route which is compatible with self-administration and dosing in space and the battlefield). However, despite significant efforts, and advances in developing radiation countermeasures, both protectors and mitigators are still an unmet need. It is therefore important to continue the search for efficient countermeasures to protect the general population in the event of nuclear terrorism or accident, as well as on the battlefield (in the event of radiological explosive devices being used), as well as astronauts and space travelers. Hence, our findings of the effective countermeasure activity of AP given as a diet supplement for Si-induced toxicity to the hematopoietic and the GI would have a significant impact on radiation protection during the space mission and on earth. Our data on the impact of the AP diet on the gut microbiota of mice with or without exposure to silicon ions would significantly have an important impact in optimizing the diet with AP, as well as integration with other bioactive compounds (e.g. other flavonoids and polyphenols), pre-biotic and pro-biotic compounds for potentially synergistic beneficial effects to protect the general population or patients who will be undergoing or have undergone radiation therapy. Further, our data on the efficacy of AP in the protection/prevention and/or mitigation of radiation-induced cancer will have a significant impact on the future development of a novel therapeutic strategy to prevent or mitigate the acute or long-term effects of radiation during a space mission, in the event of radiobiological events on earth, (e.g. a nuclear accident or terrorisms) or in the battlefield.
Task Progress:	We have completed the study using male mice as an experimental model. The results from our project demonstrate that apigenin (a flavone compound) given to mice as a diet supplement is a viable and novel radiation countermeasure for space exploration. Our results show that apigenin is a potent radioprotector and a potent radiomitigator. Our results also indicate that such beneficial effects of apigenin are based on its potent anti-inflammation and restoration of radiation-induced gut dysbiosis. In summary, the highlights of our studies are: 1. The highlights of our findings on the countermeasure effectiveness of apigenin (AP) on hematopoietic cells are: • AP prevents a loss of white blood cells (leukopenia). • AP enhances the production of red blood cells (erythropoiesis). • AP prevents depletion of platelets (thrombocytopenia). • AP protects and enhances the proliferation of hematopoietic stem/progenitor cells (HSPCs). Our data demonstrate for the first time that AP is an important countermeasure because AP can prevent leukopenia and thrombocytopenia. Currently, there are no reports of agents that can simultaneously prevent hematological abnormalities induced by radiation exposure. This information is essential for the field of radiation countermeasures. • AP prevents Si-induced acute leukemia/lymphoma in exposed mice. 2. The highlights of our findings on the countermeasure effectiveness of AP on the gut tissue are: • Consumption of AP protected mice from radiation-induced gut injuries. • Daily AP consumption restored microbiota balance in the gut tissue of irradiated mice. • Inflammation was suppressed in the gut of AP-fed mice with or without irradiation. • AP is a viable and novel countermeasure for radiation-induced gut injuries. Since there are sex differences in response to radiation and apigenin metabolism, it is important to include female mice in future studies.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Articles in Peer-reviewed Journals	Rithidech KN, Peanlikhit T, Honikel L, Li J, Liu J, Karakach T, Zimmerman T, Welsh J. "Consumption of apigenin prevents radiation-induced gut dysbiosis in male C57BL/6J mice exposed to silicon ions." Radiat Res. 2024 Apr 1;201(4):317-29. https://doi.org/10.1667/RADE-23-00110.1 ; PubMed PMID: 38373016 , Apr-2024
Articles in Peer-reviewed Journals	Peanlikhit T, Aryal U, Welsh JS, Shroyer KR, Rithidech KN. "Evaluation of the inhibitory potential of apigenin and related flavonoids on various proteins associated with human diseases using AutoDock." Int J Mol Sci. 2025 Mar 12;26(6):2548. https://doi.org/10.3390/ijms26062548 ; PMID: 40141193 , Mar-2025

Articles in Peer-reviewed Journals	Peanlikhit T, Liu J, Ahmed T, Welsh JS, Karakach T, Shroyer KR, Whorton E, Rithidech KN. "Daily consumption of apigenin prevents acute lymphoma/lymphoblastic leukemia in male C57BL/6J mice exposed to space-like radiation." Cancers. 2025 Oct 31;17(21):3513. https://www.mdpi.com/2072-6694/17/21/3513 , Oct-2025